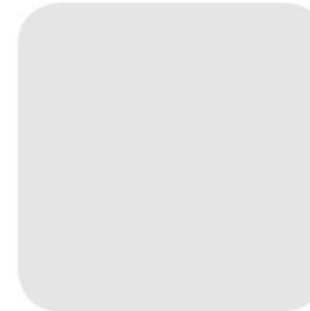
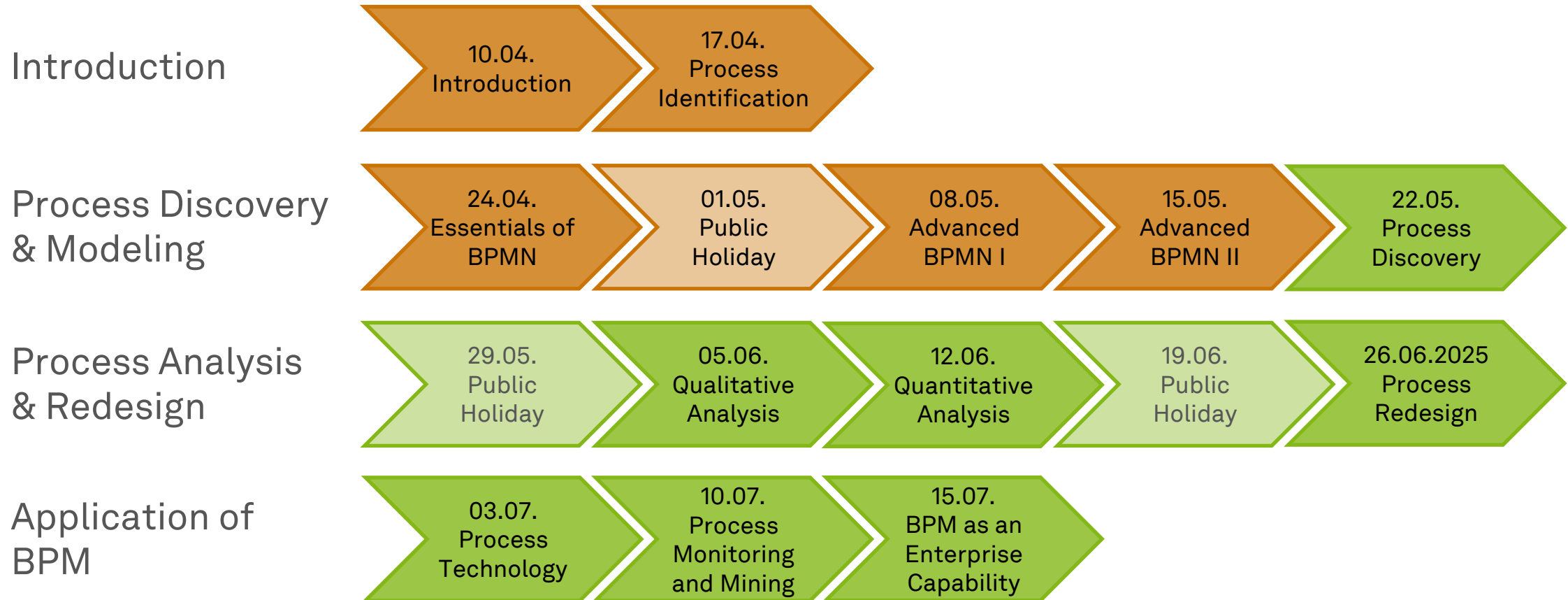


Business Process Management

Process Discovery



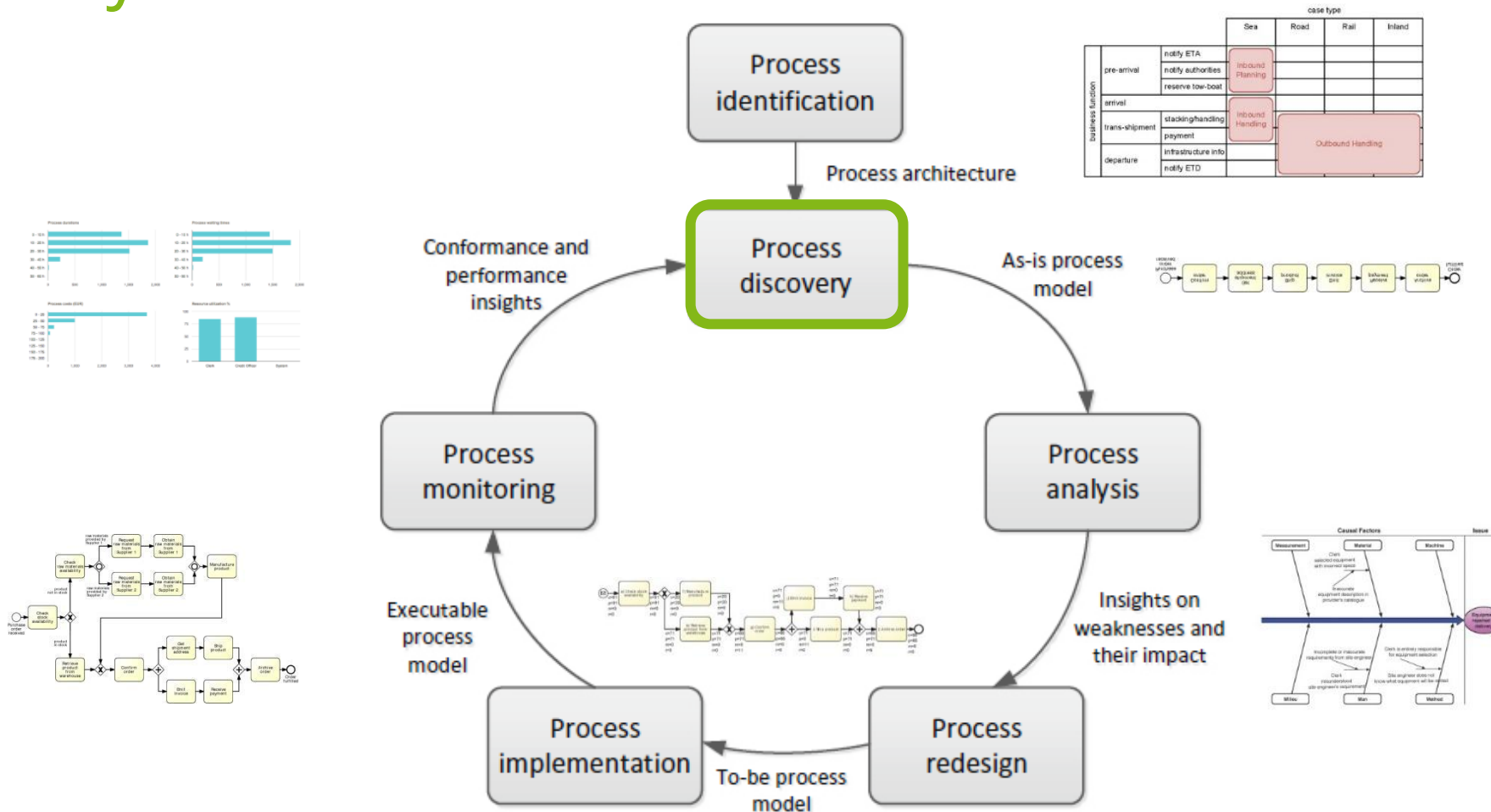
Roadmap



Learning Goals of Today

- Understand the challenges of process discovery
- Understand and be able to choose process discovery methods
- Be able to use process modeling during process discovery

BPM Lifecycle



Dumas et al. (2018)

Process Discovery is...

- ...the act of **gathering information** about an existing process and ...
- ...**organizing it** in terms of an as-is process model



→ **Process discovery is a much broader activity than just modeling a process**

Process Discovery Phases

1. Define the setting

- Assemble a team in a company that will be responsible for working on the process

2. Gather information

- Build an understanding of the process with discovery methods

3. Conduct the modeling task

- Create the process model(s)

4. Assure process model quality

- Guarantee that the resulting process models meet different quality criteria

Define the Setting: Process Discovery



Process Discovery: Who is involved?



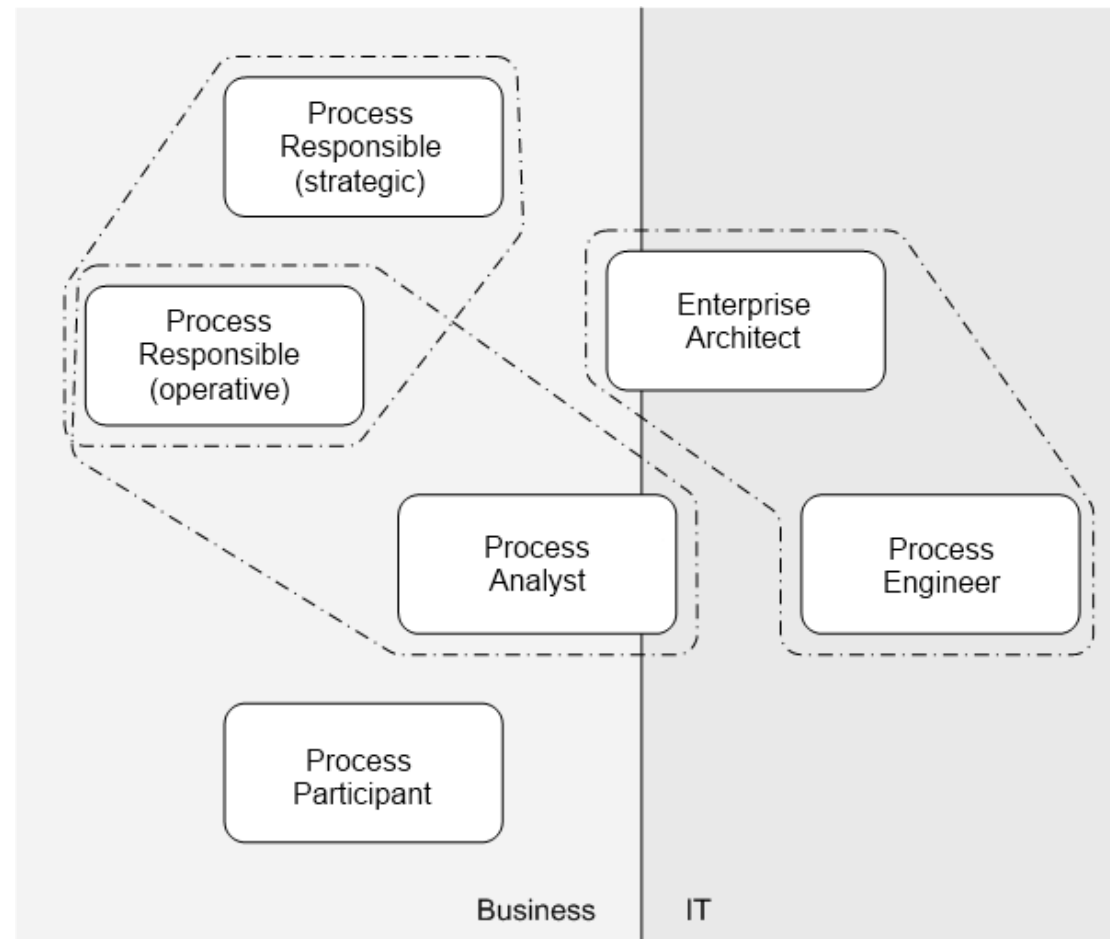
Process Analyst



Domain Expert

- **Process owner** needs to secure commitment and involvement of both

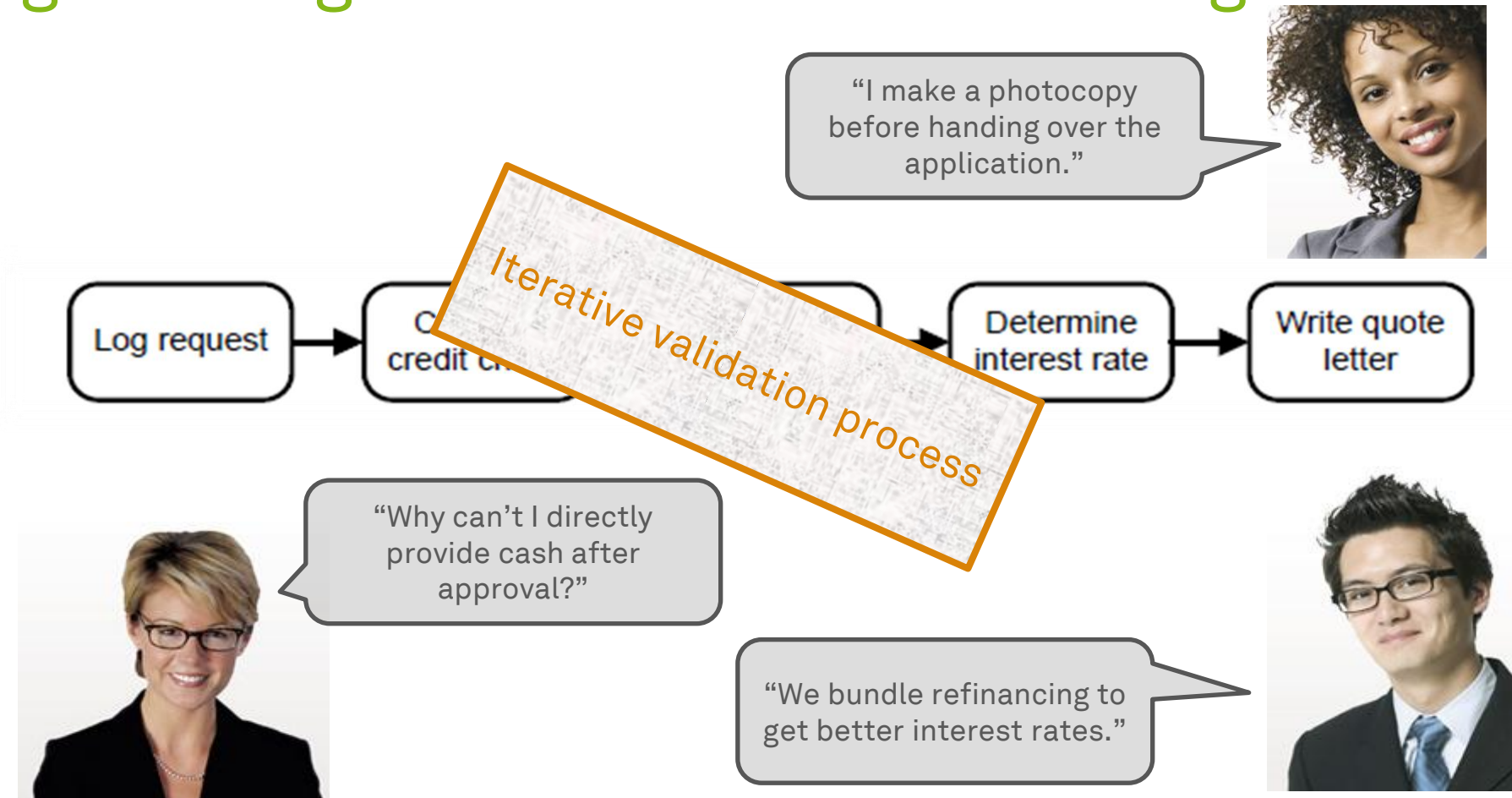
Involvement of Business and IT



Strength and Weaknesses

Aspect	Process Analyst	Domain Expert
Modeling Skills	Strong	Limited
Process Knowledge	Limited	Strong

Challenge of Fragmented Process Knowledge



Challenge of Instance/ Case Level Thinking

Welcome - Already a member? [[Sign In](#)] [[My Itinerary](#)]

Home Vacation Packages Hotels Cars Flights Cruises Things to Do DEALS

PLAN YOUR TRIP WITH EXPEDIA

☒ Flight
☐ Hotel
☐ Car
☐ Activities
☐ Cruises

Flight

☒ Roundtrip ☐ One-way

☐ My dates are flexible (popular routes)

Leaving from:

Going to:

Returning: mm/dd/yy

Adult (18-64) Seniors (65+) Children (0-17)

Show Additional Options ▾

BEST PRICE GUARANTEE

SEARCH FOR FLIGHTS

SEARCH FOR FLIGHT+HOTEL

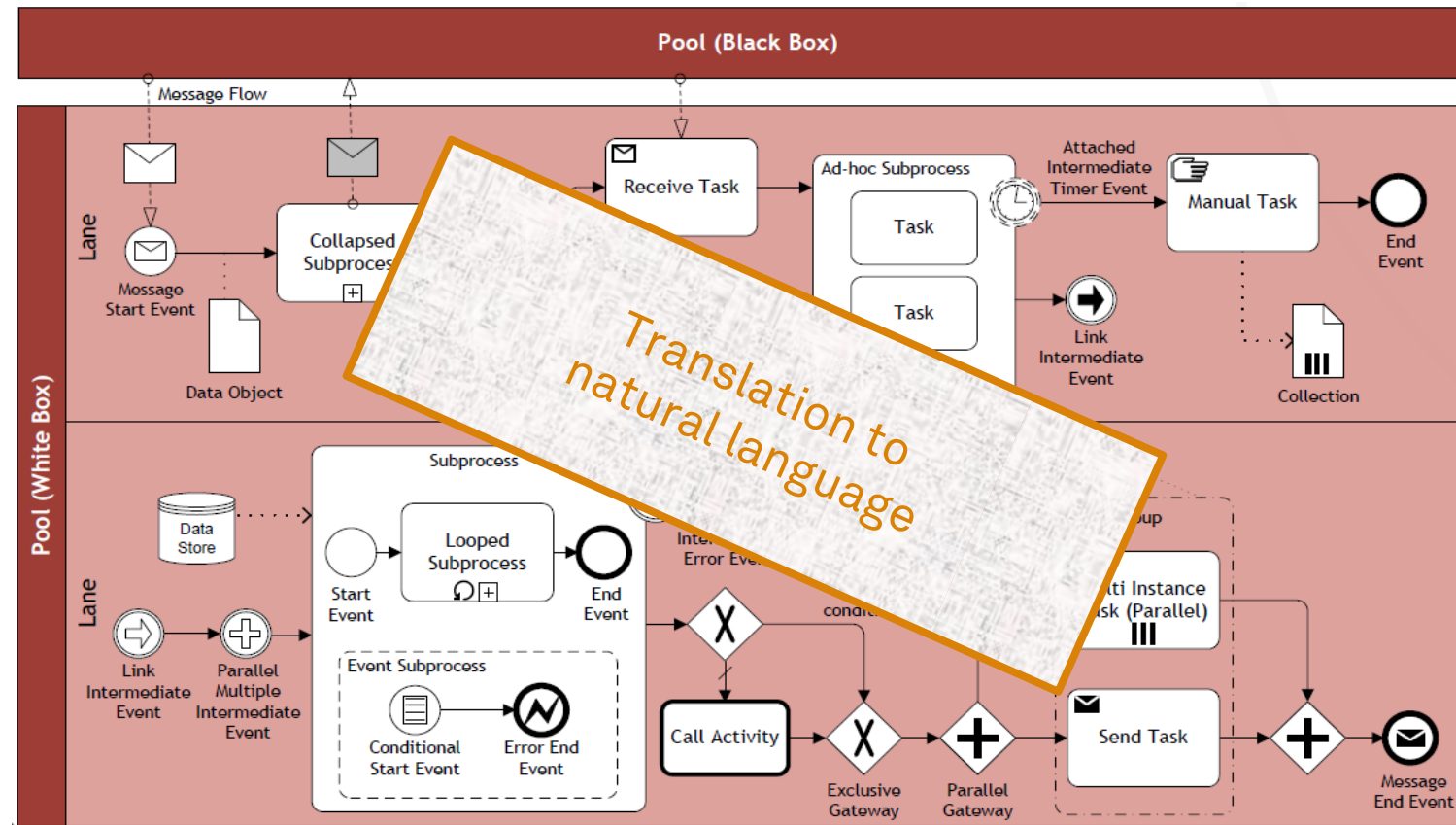
Abstraction from instance level to process level

“Every trip is different.”

“You cannot really compare. Our customers go to different places in different seasons using different modes of transportation.”

“We can never do anything exactly in the same way. There are so many special conditions.”

Challenge of Unfamiliarity with Process Modeling Methods



Profile of an Expert Process Analyst

- **Five Factor Model**
 - openness (appreciating art, emotion, and adventure)
 - conscientiousness (tendency to self-discipline, achievement, and planning)
 - extraversion (being positive, energetic, and seeking company)
 - agreeableness (being compassionate and cooperative)
 - neuroticism (being anxious, depressed, and vulnerable)

Profile of an Expert Process Analyst

- Five Factor Model for Analysts
 - openness (appreciating art, emotion, and adventure)
 - **conscientiousness** (tendency to self-discipline, achievement, and planning)
 - **extraversion** (being positive, energetic, and seeking company)
 - agreeableness (being compassionate and cooperative)
 - neuroticism (being anxious, depressed, and vulnerable)

The Problem with Process Discovery

- Process discovery in general belongs to the category of **ill-defined problems**
- This means in the beginning of a process discovery project, it is **not** exactly **clear**
 - who of the **domain experts** must be contacted,
 - which **documentation** can be used, and
 - which **agenda** the different stakeholders might have in mind

Process Discovery Guidelines

- **Get the right people on board**
 - When talking to a participant make sure his supervisor is involved
- **Have a set of working hypotheses**
 - Prepare questions and assumption for workshops to structure the process at different levels of detail
- **Identify patterns**
 - Gather facts on alternatives and conditions, independence and concurrency of activities to sketch process models
- **Pay attention to model quality**
 - Presentation is important
 - Use the right level of abstraction and the right set of elements

Gather Information: Process Discovery Methods



Process Discovery Methods

- **Evidence-based**
 - Document analysis
 - Observation
 - Automated process discovery
- **Dialog-based**
 - Interview-based
 - Workshop-based

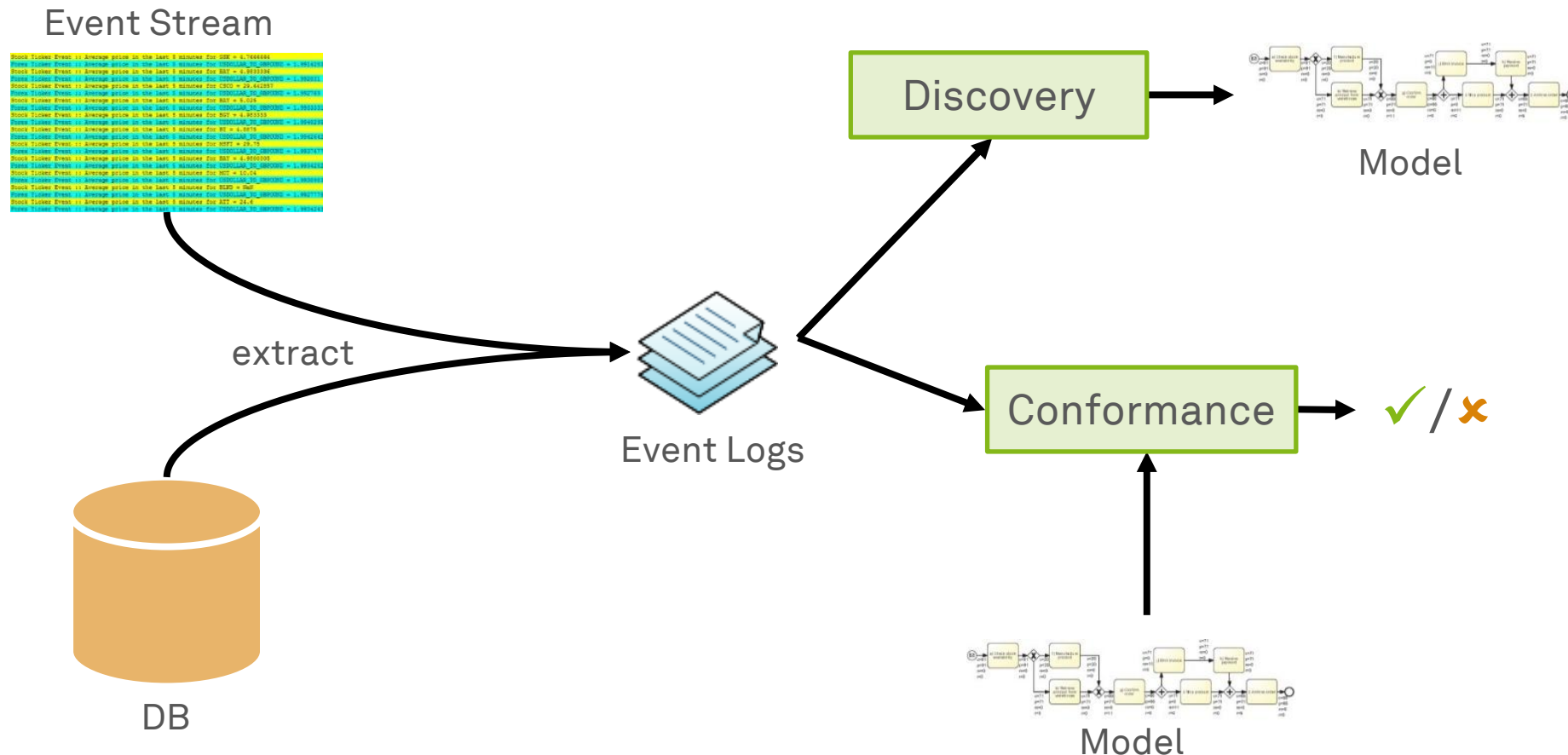
Document Analysis

- Documents point to existing **roles**, **activities**, and **business objects** and their **relations**
 - Process descriptions (ideal scenario) 😊
 - Internal policies
 - Organization charts & employment plans
 - Quality certificate reports
 - Glossaries and handbooks
 - Forms
 - Work instructions
- Potential Issues
 - Document **organization might not be process-oriented**
 - Document **granularity might not fit**
 - Documents **might not show reality**
- This method is helpful **before** talking to domain experts

Observation

- Follow directly the processing of individual cases
 - **Active role**
 - play a specific role, e.g. customer
 - **Passive role**
 - observe participants and their environment (shadowing)
- Trace business objects in the course of their lifecycle
- Inspect the work environment
- Potential Issues
 - Active role: **cases only**, no big picture
 - Passive role: participants' **bias** when being observed
- This method reveals how a process is conducted in reality as of today

Automated Process Discovery

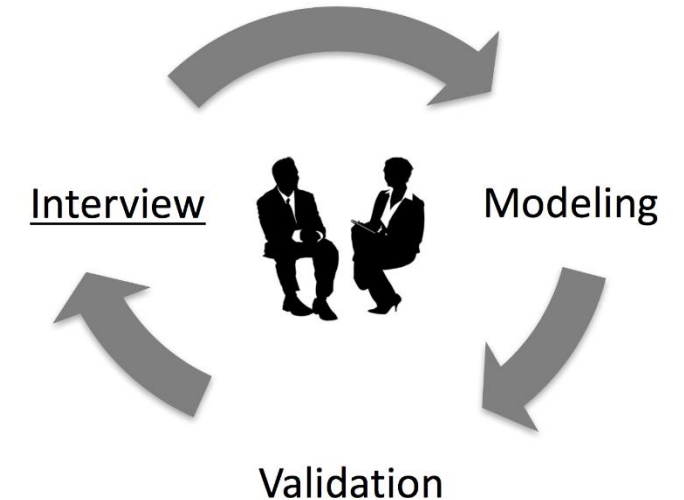


Automated Process Discovery

- Event logs capture actual execution
 - **Objective** representation of the process
- **Rich set** of process-related information beyond the tasks
 - e.g., timestamps, resources, business objects
- **Not limited** to one system
 - End-to-end-process representation
- Potential Issues
 - (Quality) event logs may not be **not always available**
 - Event logs **may be fragmented** (granularity, missing values)
 - Event logs **only represent the interaction with the system** (see manual tasks)
- This method reveals how a process is executed with BPM systems

Interviews

- **Procedure**
 - Interview – Model – Validate // Iterate
- **Strategies**
 - **Backward** (upstream) vs. **forward** (downstream) perspective
 - **Structured** vs. **unstructured** (free-form) interviews
 - Recommendation: $\frac{3}{4}$ to $\frac{1}{4}$ ratio
 - “**Sunny day**“ vs. “**rainy day**“
- Remember the **mental model** (e.g., shared terminology)
- This method reveals a rich picture of the process and its participants



Workshop

- Gathers **all key stakeholders** together
- Participants interact to create **shared understanding**
- Typically, one **process analyst** (facilitator), multiple **domain experts** (modeler, scribe), **process owner** may also attend
- **Model** is directly created during the workshop (typically a separate role – tool operator)
- Usually 3 to 5 half-day sessions

Potential Issues

- Team opinion **may not be the sum** of individual opinions
- **Not everybody dares to speak up**



- This method reveals a rich and consolidated picture of the process and its participants

Dumas et al. (2018)



Can I apply any method in any organization?



Company Culture

- Before starting with process discovery, understand the **culture and the sentiment** of an organization
- Companies can preach and practice an **open culture** in which all employees are encouraged to utter their ideas and their criticism
 - They can benefit a lot from workshops as participants are likely to present their ideas freely
- **Strictly hierarchical** organizations can be close-mouthed
 - It is necessary to take special care that every participant gets an equal share of attention
 - Ideas and critique are often held back

Method Results

Aspect	Evidence	Interview	Workshop
Objectivity	high	medium-high	medium-high
Richness	medium	high	high
Time consumption	low-medium	medium	medium
Immediacy of feedback	low	high	high

Method Strengths and Weaknesses

Technique	Strength	Weakness
Document Analysis	✓ Structured information ✓ Independent from availability of stakeholders	× Outdated material × Wrong level of abstraction
Observation	✓ Context-rich insight into process	× Potentially intrusive × Stakeholders likely to behave differently × Only few cases and not all processes can be observed
Automatic Discovery	✓ Extensive set of cases ✓ Objective data	× Potential issue with data quality and level of abstraction × Data may not be available or be available only in part × Data extraction and preparation is time-consuming
Interview	✓ Context-rich insights	× Requires time of process stakeholders × Time-consuming: several iterations required before sign-off
Workshop	✓ Context-rich insights ✓ Direct resolution of conflicting views	× Synchronous availability of several stakeholders × Time-consuming: multiple sessions typically required

Conduct the Modeling Task: Process Modeling in Process Discovery



Modeling the Process

1. Identify the **process boundaries**
2. Identify **activities** and **events**
3. Identify **resources** and their **handoffs**
4. Identify the **control flow**
5. Identify **additional elements**

1) Identify the Process Boundaries

- What are the **process triggers**?
 - Manual, timer, message, ...
- What are the possible **outcomes** ?
 - Positive, negative
- Which **perspective** do we assume?
 - Internal, external, collaborative
- What **business objects** (artifacts) are required to the process?
 - Input, output

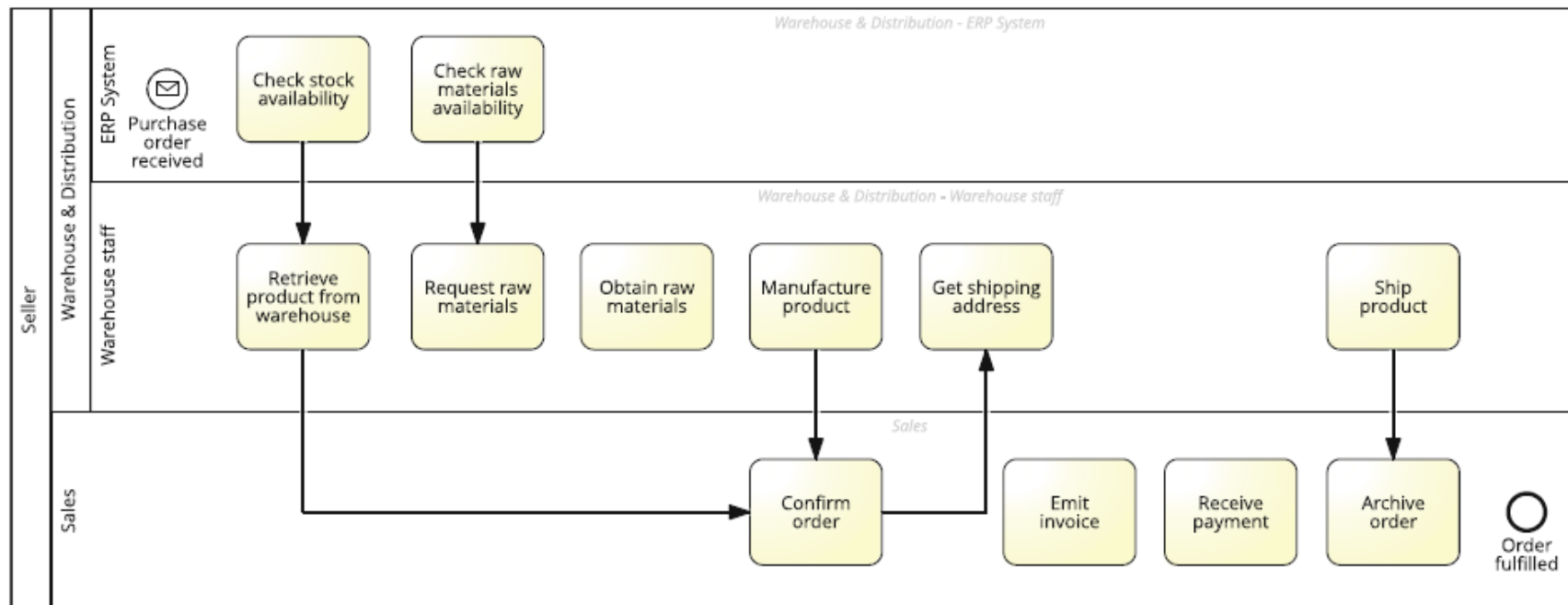
2) Identify Activities and Events

- Which are the **main activities**?
- What are **significant events**?

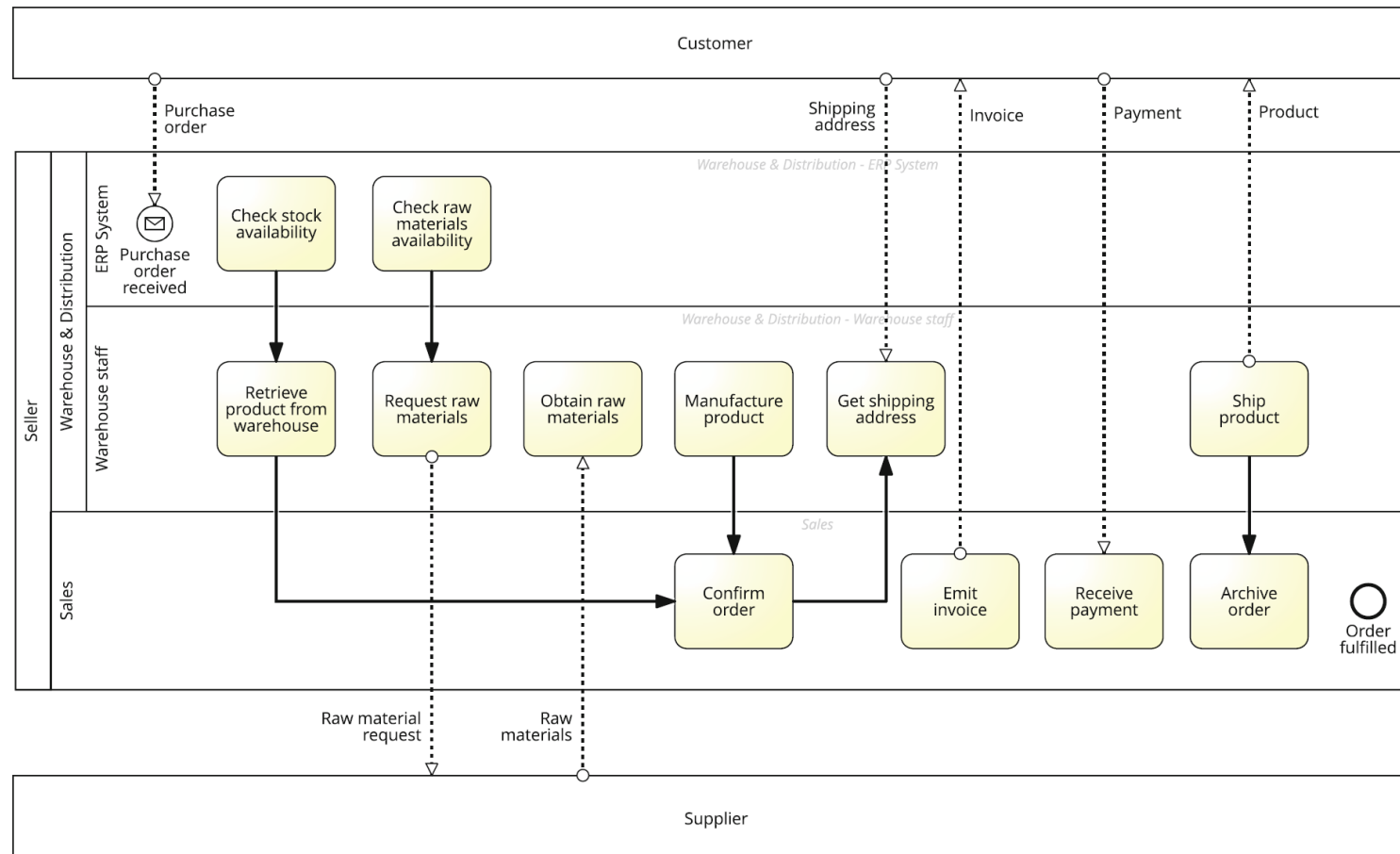


3) Identify Resources and their Handoffs

- Who is **responsible**?
- Where do we find **handoffs**?

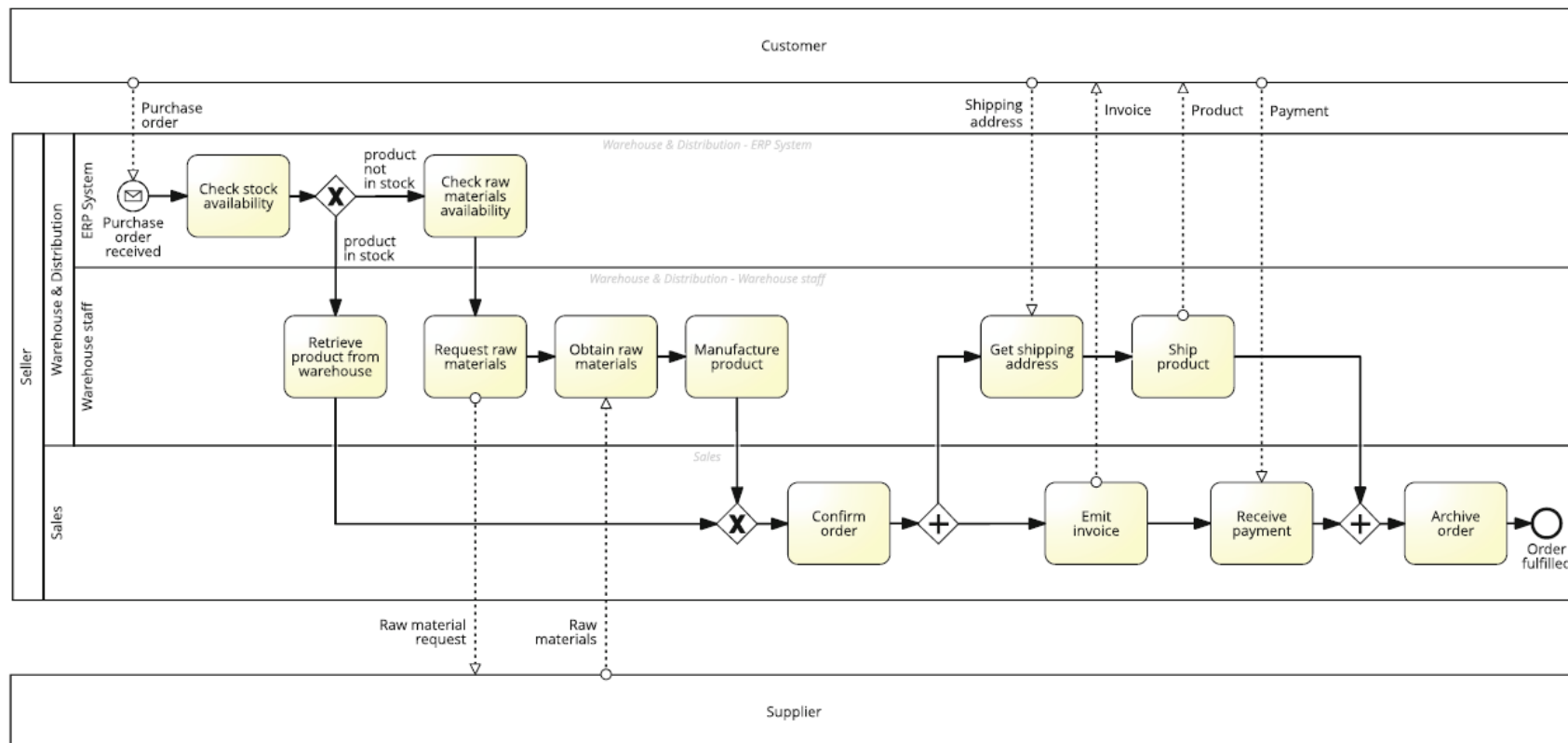


3) Identify Resources and their Handoffs



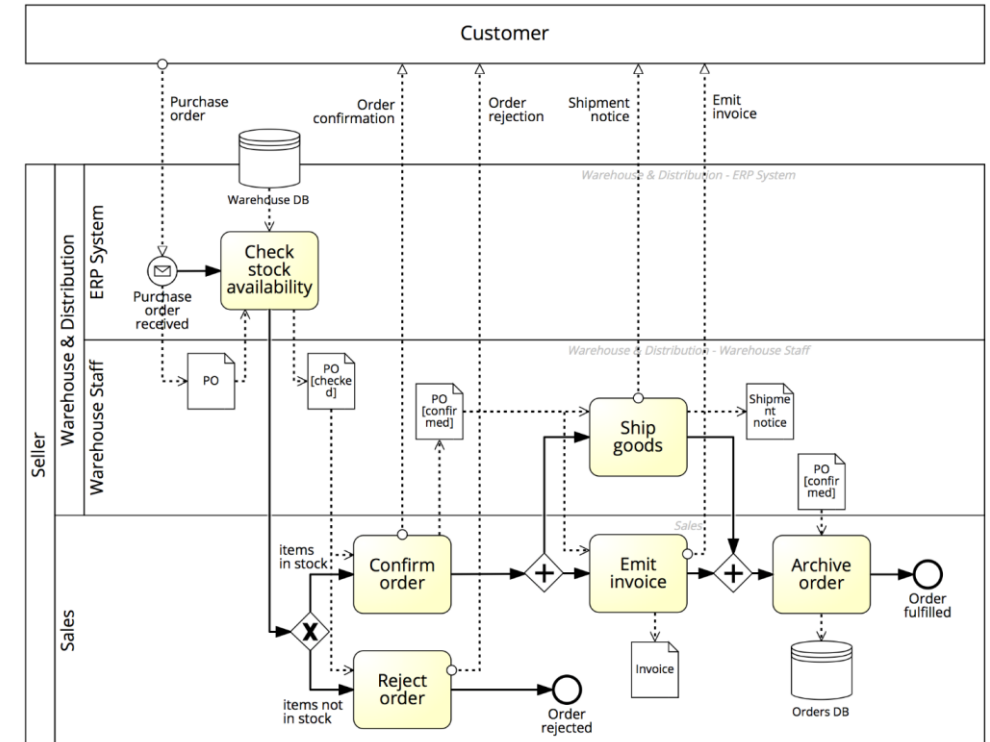
4) Identify the Control Flow

- When and why are activities executed?



5) Identify Additional Elements

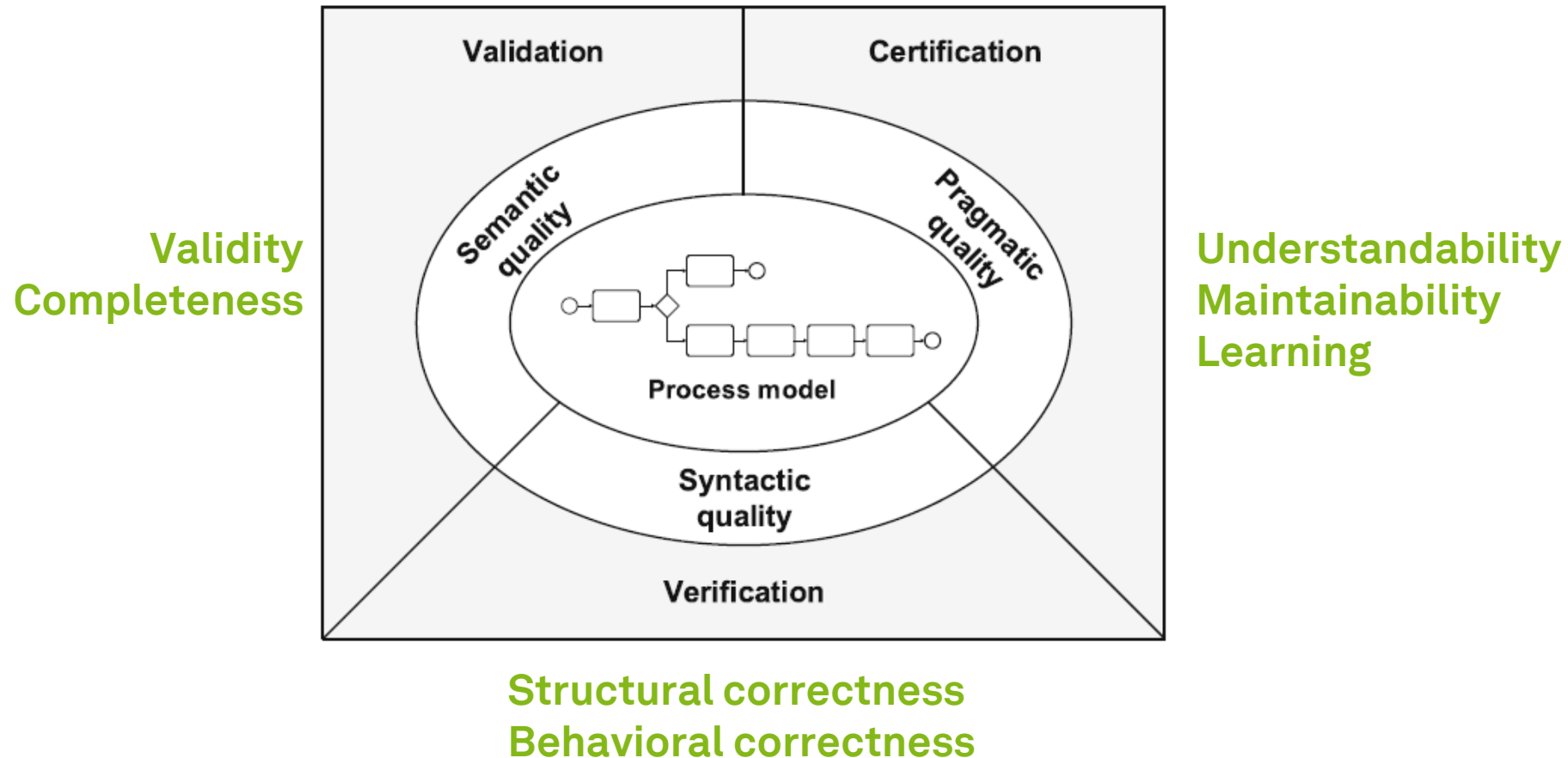
- What are involved **business objects**?
 - Data objects
 - Data stores
- What are involved **exception handlers**?
 - Boundary events
 - Exception flows
 - Compensation handlers
- What **data** is necessary for **execution**?
- Add risk or cost data
- **When do we stop modeling?**



Assure Process Model Quality: Process Models in Process Discovery

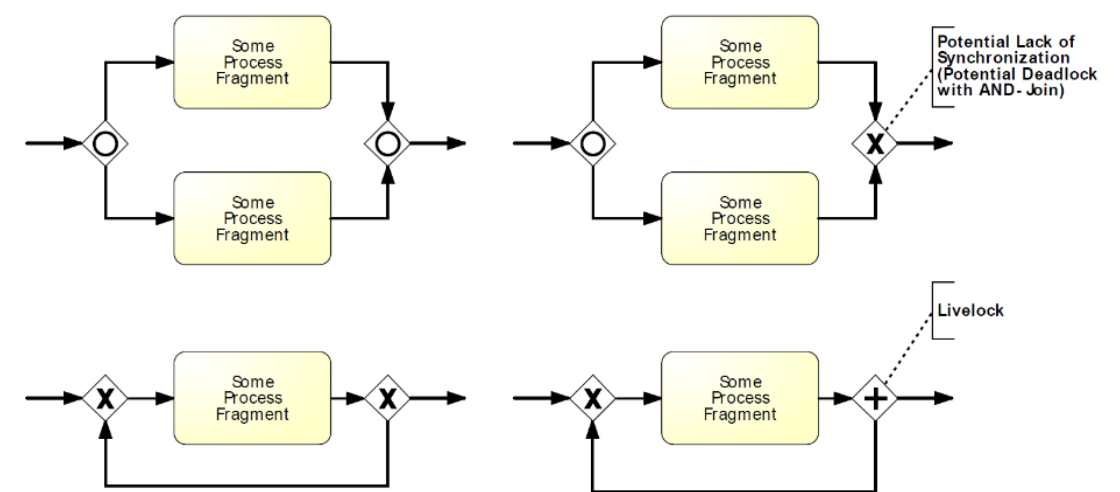
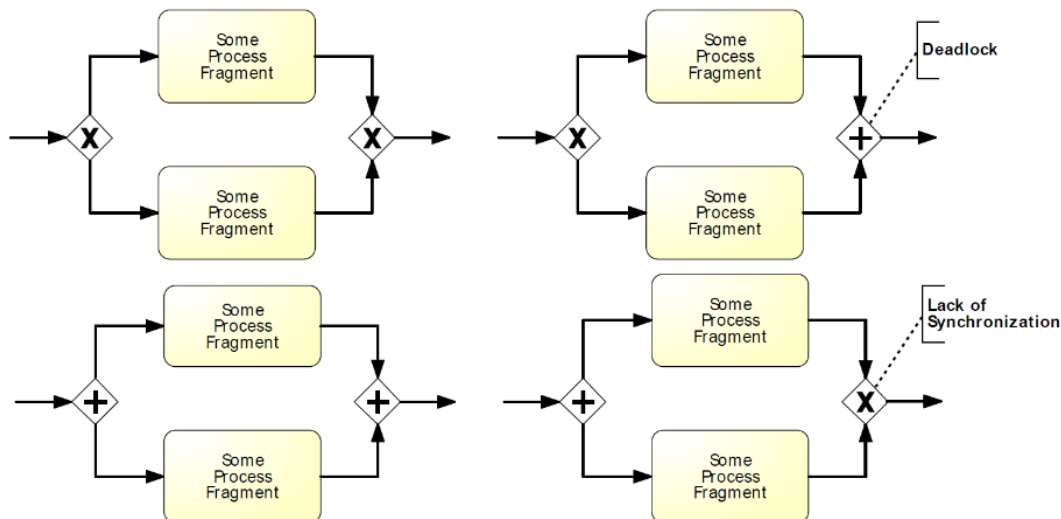


Process Model Quality Assurance



Verification: Examples of Behavioral Anomalies

- Sound Model
 - **Option to complete** (any instance must complete)
 - **Proper completion** (each remaining token must be in a different end event)
 - **No dead activities** (any activity can be executed)



Dumas et al. (2018)

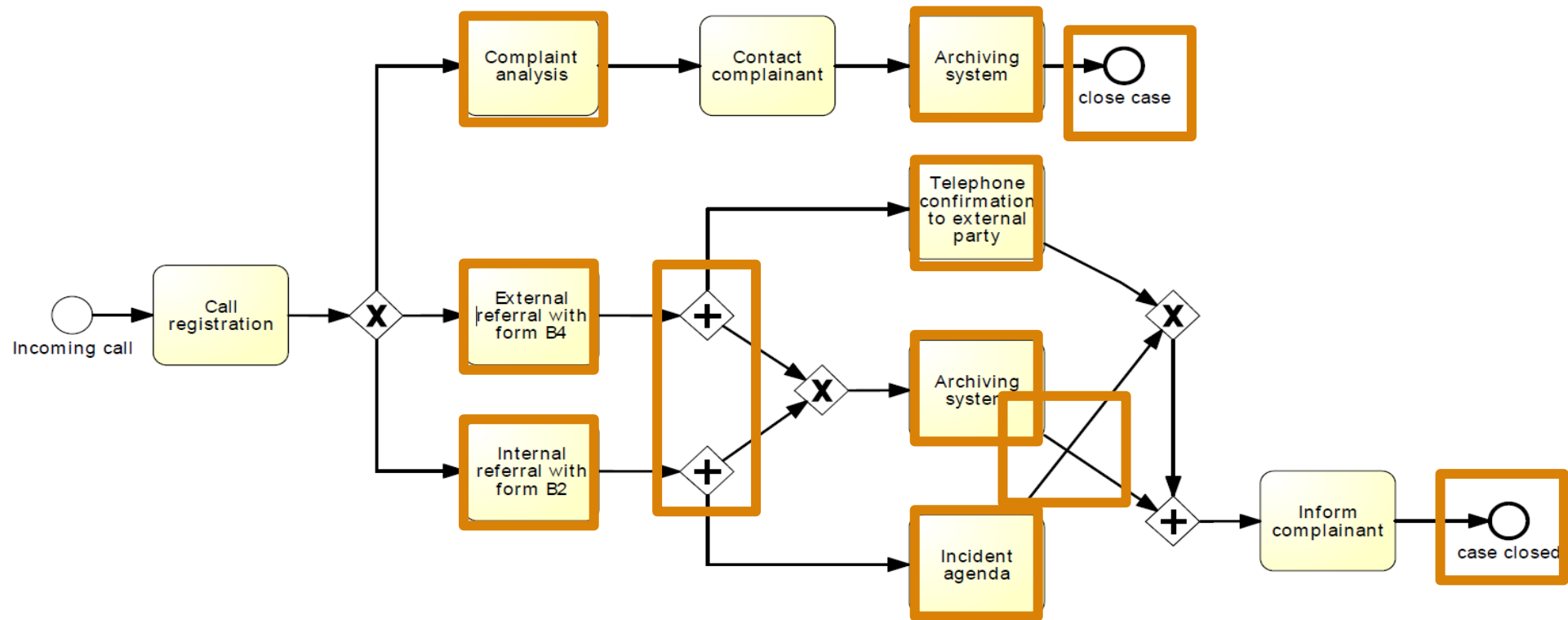
Process Modeling Guidelines and Conventions

- Objectives
 - Safeguard **model consistency** and improve **standardization** and reuse
 - Reduce the **dependency** on individual process analysts
 - Facilitate **access** to models by non-modeling experts
- Restrictions
 - **Vocabulary** (avoid certain elements, e.g. subprocesses)
 - **Structure** (limit the structure of the model, e.g. by restricting layers)
 - **Semantics** (avoid element meanings, e.g. limiting boundary events to business faults)
 - **Appearance** (restrict the model appearance, e.g. labels, layout, and notation)

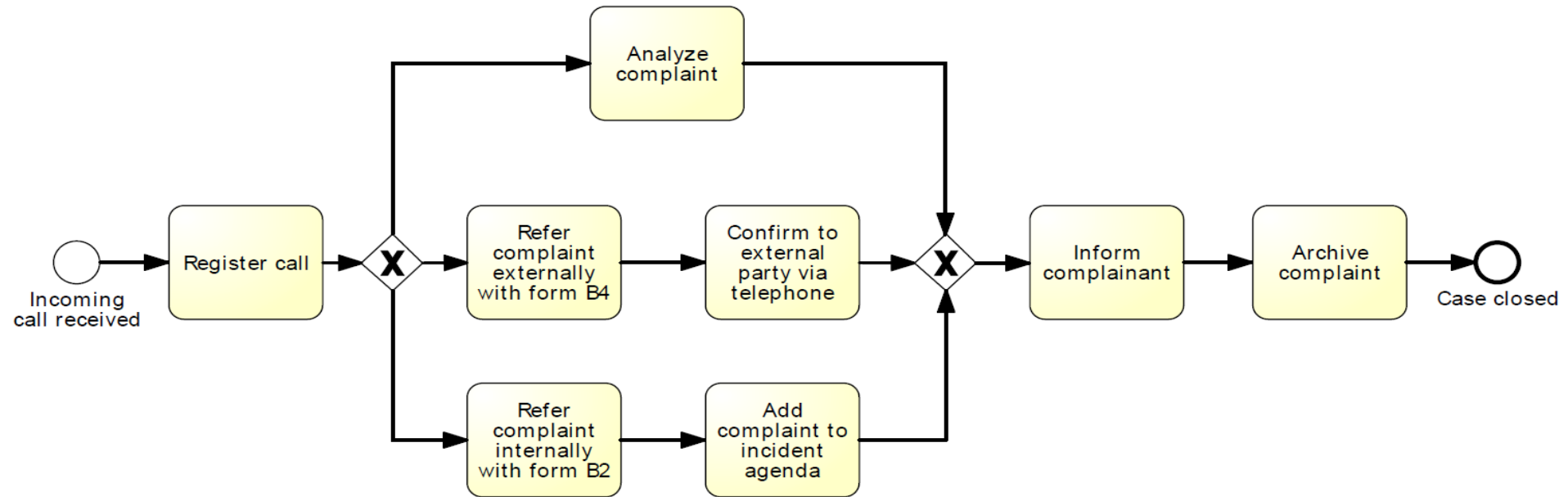
Seven Process Modeling Guidelines (7PMG)

- G1** Use as few elements in the model as possible
- G2** Minimize the routing paths per element
- G3** Use one start and one end event
- G4** Model as structured as possible
- G5** Avoid, where possible, OR routing elements
- G6** Use verb-object activity labels
- G7** Decompose a model with more than 30 elements

Bad Example



Good Example



Learning Goals of Today

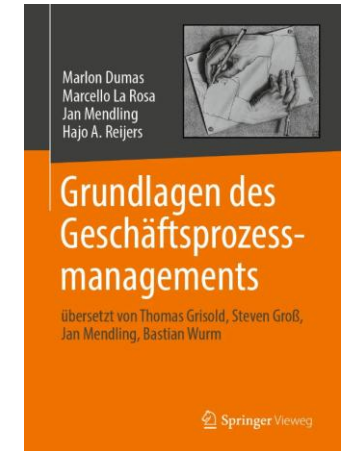
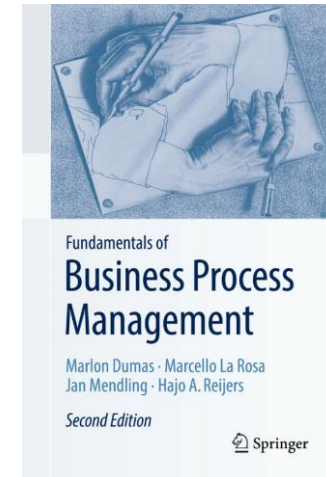
- Understand the challenges of process discovery
- Understand and be able to choose process discovery methods
- Be able to use process modeling during process discovery

Recap Questions

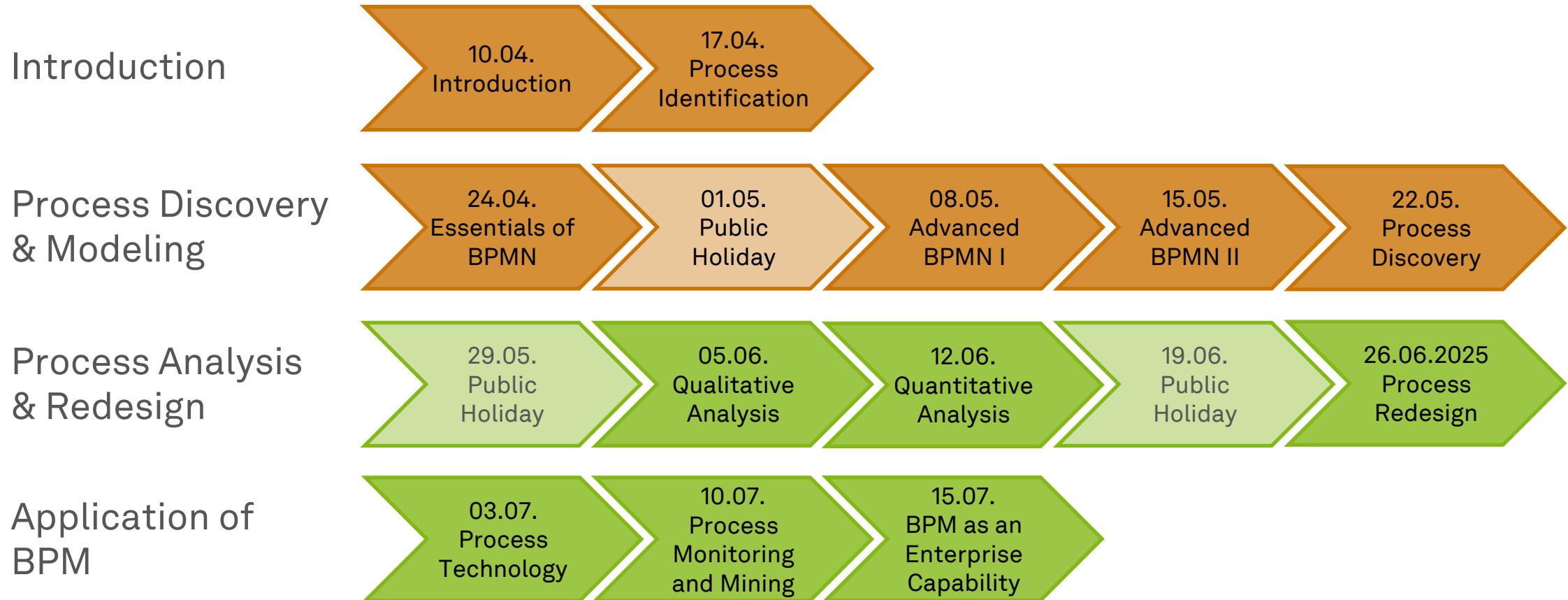
- What is process discovery and what are its four phases about?
- Which evidence-based and dialogue-based process discovery methods do you know, what are their benefits and drawbacks, and when can they be applied?
- Which dimensions for process model quality assurance are you aware of?

Literature

- **Section 4: Process Discovery**



Roadmap



Any Questions about Process Discovery?





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